

Notes on Local $\mathbb{F}_0$ and $\mathbb{F}_1$

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Abstract

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1. Local $\mathbb{F}_0$

1.1 Mirror curve

We write the mirror curve to $\mathbb{F}_0$ as the following embedding in $\mathbb{C}^* \times \mathbb{C}^* \ni (w, X)$

$$w + \frac{1}{w} + \frac{1}{R^2} \left(X + \frac{1}{X} + U \right) = 0. \quad (1.1)$$

We can write this curve as a quartic by introducing $Y = -R^2 X \left(w - \frac{1}{w} \right)$,

$$Y^2 = f_4(X) = (X^2 + UX + 1)^2 - 4X^2 R^2. \quad (1.2)$$

The discriminant δ turns out to be (upto a numerical constant)

$$\delta = R^8 \Delta(U, R), \quad \Delta(U, R) = 16R^8 - 8R^4 (U^2 + 4) + (U^2 - 4)^2. \quad (1.3)$$

The equation $\Delta(U, R) = 0$ has four solutions

$$U_1 = 2 - 2R^2, \quad U_2 = -2 + 2R^2, \quad U_3 = -2 - 2R^2, \quad U_4 = 2 + 2R^2. \quad (1.4)$$

In the real $U - R$ slice, the condition is plotted in fig. 1.

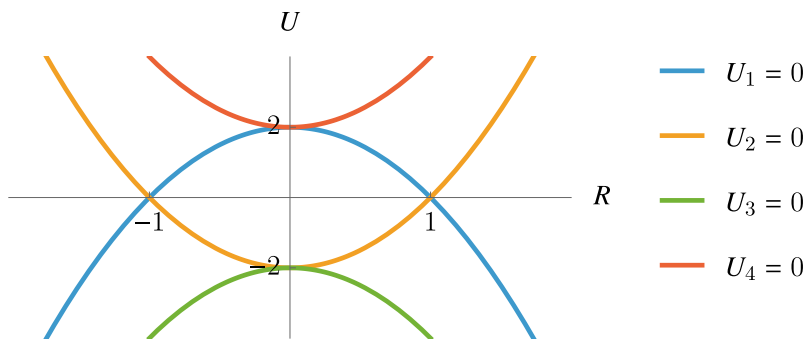


Figure 1: The locus $\Delta(U, R) = 0$

We can turn the quartic in eq. (1.2) to cubic form by the following rational transformation

$$X = \frac{ax + b}{cx + d}, \quad Y = \frac{ad - bc}{(cx + d)^2} y. \quad (1.5)$$

The curve now looks like

$$Y^2 = \frac{(ad - bc)^2}{(cx + d)^4} y^2 = f_4 \left(\frac{ax + b}{cx + d} \right). \quad (1.6)$$

This implies,

$$(ad - bc)^2 y^2 = f_4 \left(\frac{ax + b}{cx + d} \right) (cx + d)^4 = c^4 f_4 \left(\frac{a}{c} \right) x^4 + f_3(x), \quad (1.7)$$

where f_3 is a cubic polynomial in x whose coefficient of x^3 is directly proportional to $f_4'(a/c)$. It follows that if we set a/c to be a simple root of f_4 and $ad - bc = 1$ then we obtain the

desired cubic form. We can then scale and translate so as to get the cubic in the usual Weierstrass form: $y^2 = 4x^3 - g_2x - g_3$.

Performing this procedure on the quartic in eq. (1.2), one gets

$$\begin{aligned} g_2 &= \frac{1}{12} \left(16R^8 - 8R^4 (U^2 + 2) + (U^2 - 4)^2 \right), \\ g_3 &= \frac{1}{216} \left(4R^4 - U^2 + 4 \right) \left(16R^8 - 8R^4 (U^2 + 5) + (U^2 - 4)^2 \right). \end{aligned} \quad (1.8)$$

We also find the j -invariant defined as $j = \frac{g_2^3}{g_2^3 - g_3^2}$,

$$j = \frac{\left(16R^8 - 8R^4 (U^2 + 2) + (U^2 - 4)^2 \right)^3}{R^8 \left(16R^8 - 8R^4 (U^2 + 4) + (U^2 - 4)^2 \right)}. \quad (1.9)$$

1.2 Periods as double Laurent series

From the GLSM charges defining the toric geometry,

$$\begin{array}{ccc|cc} 0 & 0 & 1 & -2 & -2 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{array} \quad (1.10)$$

we see that the periods of the mirror curve are given by the double sum

$$\begin{aligned} \omega_1 &= \sum_{k, \ell \geq 0} \frac{(2k + 2\ell)!}{(k!)^2 (\ell!)^2} z^{k+\ell} \lambda^\ell = F_4\left(\frac{1}{2}, 1; 1, 1; 4z, 4z\lambda\right) \\ \omega_2 &= (2 \log z + \log \lambda) \omega_1 - 2 \sum_{k+\ell > 0} \frac{(2k + 2\ell)!}{(k!)^2 (\ell!)^2} (H_k + H_\ell - 2H_{2k+2\ell}) z^{k+\ell} \lambda^\ell \end{aligned} \quad (1.11)$$

where F_4 is the Appell hypergeometric series, and $H_n = \sum_{k=1}^n (1/k)$ are the harmonic numbers. For $\lambda = 1$, we recover [1, A.39],

$$\omega_1 = {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 16z\right), \quad \omega_2 = -2\pi {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; 1 - 16z\right), \quad (1.12)$$

Similarly, for $\lambda = -1$, we recover [1, A.46],

$$\omega_1 = {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; -64z^2\right), \quad \omega_2 = -\pi\sqrt{2} {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; 1 + 64z^2\right), \quad (1.13)$$

The CY periods satisfy $z\partial_z \omega_2 = \omega_1, z\partial_z \omega_3 = \omega_2$. Hence they are obtained by integrating once

with respect to z ,

$$\begin{aligned}\varpi_2 &= \log z + 2 \sum_{k+\ell>0} \frac{(2k+2\ell-1)!}{(k!)^2(\ell!)^2} z^{k+\ell} \lambda^\ell \\ \varpi_3 &= -\log^2 z + (2\log z + \log \lambda) \varpi_2 - 2 \sum_{k+\ell>0} \frac{(2k+2\ell)!}{(k!)^2(\ell!)^2(k+\ell)} \left(H_k + H_\ell - 2H_{2k+2\ell} + \frac{1}{k+\ell} \right) z^{k+\ell} \lambda^\ell\end{aligned}\quad (1.14)$$

1.3 Kerr-Doran formula for the conifold period

In [2], a formula is provided for periods of nodal curves in toric ambient spaces. Here we check that this formula reproduces the result for the period at the conifold point derived in [A.84,A.107][1],

$$\begin{aligned}T(\tau = 0, m) &= \frac{2}{\pi^2} \left(\text{Li}_2(\text{i}e^{\text{i}\pi m/2}) - \text{Li}_2(-\text{i}e^{\text{i}\pi m/2}) \right) \\ T(\tau = \frac{1}{2}, m) &= \frac{2}{\pi^2} \left(\text{Li}_2(e^{\text{i}\pi m/2}) - \text{Li}_2(-e^{\text{i}\pi m/2}) \right)\end{aligned}\quad (1.15)$$

where $R = e^{\text{i}\pi m/2}$, such that $R^4 = \lambda = e^{2\pi \text{i} m}$.

1.3.1 Rational parametrization of the nodal curve

As shown in eq. (1.4), the curve in eq. (1.2) develops four distinct branches of nodes. These correspond to points in parameter space where the elliptic curve degenerates. Following the procedure outlined in [2], we rationally parametrize each nodal curve using a parameter z , such that the limits $z \rightarrow 0$ and $z \rightarrow \infty$ map to the node of the curve.

The quartic polynomial in eq. (1.2) takes the following forms along the four branches, each associated with a corresponding node:

$$\begin{aligned}U_1 &= 2 - 2R^2, & f_4(X) &= (X+1)^2 \left(X^2 + (2 - 4R^2)X + 1 \right), & X_0 &= -1, \\ U_2 &= -2 + 2R^2, & f_4(X) &= (X-1)^2 \left(X^2 + (4R^2 - 2)X + 1 \right), & X_0 &= 1, \\ U_3 &= -2 - 2R^2, & f_4(X) &= (X-1)^2 \left(X^2 + (-4R^2 - 2)X + 1 \right), & X_0 &= 1, \\ U_4 &= 2 + 2R^2, & f_4(X) &= (X+1)^2 \left(X^2 + (4R^2 + 2)X + 1 \right), & X_0 &= -1.\end{aligned}\quad (1.16)$$

To obtain a rational parametrization for the first branch (U_1), it suffices to find a rational expression for X such that $(X^2 + (2 - 4R^2)X + 1)$ becomes a perfect square. One convenient choice is

$$X = \frac{2(-2R^2 - t + 1)}{t^2 - 1}, \quad \Rightarrow \quad \left(X^2 + (2 - 4R^2)X + 1 \right) = \left(\frac{(4R^2 - 2)t + t^2 + 1}{t^2 - 1} \right)^2. \quad (1.17)$$

The remaining parametrizations follow from the transformations $R^2 \rightarrow -R^2$ and $X \rightarrow -X$:

$$\begin{aligned}
 X_1 &= \frac{2(-2R^2 - t + 1)}{t^2 - 1}, & Y_1 &= \frac{-16R^4t + 4R^2(t-1)^3 + (t-1)^4}{(t^2 - 1)^2}, \\
 X_2 &= -\frac{2(-2R^2 - t + 1)}{t^2 - 1}, & Y_2 &= \frac{-16R^4t + 4R^2(t-1)^3 + (t-1)^4}{(t^2 - 1)^2}, \\
 X_3 &= -\frac{2(2R^2 - t + 1)}{t^2 - 1}, & Y_3 &= \frac{-16R^4t - 4R^2(t-1)^3 + (t-1)^4}{(t^2 - 1)^2}, \\
 X_4 &= \frac{2(2R^2 - t + 1)}{t^2 - 1}, & Y_4 &= \frac{-16R^4t - 4R^2(t-1)^3 + (t-1)^4}{(t^2 - 1)^2}.
 \end{aligned} \tag{1.18}$$

Each pair (X_i, Y_i) satisfies eq. (1.2) for the corresponding value of U_i .

Returning to the original toric coordinates (X, w) , the parametrizations become

$$\begin{aligned}
 w_1 &= \frac{(t-1)(2R^2 + t - 1)}{2R^2(t+1)}, & w_2 &= -\frac{(t-1)(2R^2 + t - 1)}{2R^2(t+1)}, \\
 w_3 &= \frac{(t-1)(2R^2 - t + 1)}{2R^2(t+1)}, & w_4 &= -\frac{(t-1)(2R^2 - t + 1)}{2R^2(t+1)}.
 \end{aligned} \tag{1.19}$$

This yields a rational parametrization of the nodal curve in toric coordinates, though the node still lies at a finite distance rather than at 0 or ∞ . To remedy this, we perform one final change of variables. For U_1 , the node at $X_0 = -1$ corresponds to $t = 1 \pm 2R$, so we define

$$z = \frac{t - (1 - 2R)}{t - (1 + 2R)}, \tag{1.20}$$

similarly, for U_2 . For U_3 and U_4 , the node corresponds to $t = 1 \pm 2iR$, so we define

$$z = \frac{t - (1 - 2iR)}{t - (1 + 2iR)}. \tag{1.21}$$

Therefore, for each branch we get X and w in the z -parametrization:

$$\begin{aligned}
 X_1 &= -\frac{(z-1)(R(z-1) + z + 1)}{(z+1)(Rz + R + z - 1)}, & w_1 &= \frac{(z+1)(R(z-1) + z + 1)}{(z-1)(Rz + R + z - 1)}, \\
 X_2 &= \frac{(z-1)(R(z-1) + z + 1)}{(z+1)(Rz + R + z - 1)}, & w_2 &= -\frac{(z+1)(R(z-1) + z + 1)}{(z-1)(Rz + R + z - 1)}, \\
 X_3 &= \frac{(z-1)(R(z-1) + i(z+1))}{(z+1)(R(z+1) + i(z-1))}, & w_3 &= \frac{(z+1)(R(z-1) + i(z+1))}{(z-1)(R(z+1) + i(z-1))}, \\
 X_4 &= -\frac{(z-1)(R(z-1) + i(z+1))}{(z+1)(R(z+1) + i(z-1))}, & w_4 &= -\frac{(z+1)(R(z-1) + i(z+1))}{(z-1)(R(z+1) + i(z-1))}.
 \end{aligned} \tag{1.22}$$

1.3.2 Conifold period for a real curve

Applying the method of [2, Section 4.2], we find for the first branch the exponents and roots of the linear factors (with $i = 1, j = 1, \dots, 4$):

$$\begin{aligned} (d_j) &= (1, 1, -1, -1), & (\alpha_j) &= \left(1, \frac{R-1}{R+1}, -1, \frac{1-R}{R+1}\right), \\ (e_j) &= (1, 1, -1, -1), & (\beta_j) &= \left(-1, \frac{R-1}{R+1}, 1, \frac{1-R}{R+1}\right). \end{aligned} \quad (1.23)$$

The (imaginary part of the) non-vanishing period at the conifold point is then

$$\mathcal{P}(R) = \frac{1}{2\pi} \sum_{j,k} d_j e_k D_2\left(\frac{\alpha_j}{\beta_k}\right), \quad (1.24)$$

where $D_2(z)$ is the Bloch–Wigner function,

$$D_2(z) = \text{Im}(\text{Li}_2(z)) + \arg(1-z) \log|z|, \quad (1.25)$$

and $\text{Li}_2(z)$ denotes the ordinary dilogarithm [3].

We thus find, up to an overall sign,

$$\mathcal{P}(R) = \frac{1}{\pi} \left[-D_2\left(\frac{R+1}{1-R}\right) + D_2\left(\frac{R+1}{R-1}\right) + D_2\left(\frac{1-R}{R+1}\right) - D_2\left(\frac{R-1}{R+1}\right) \right]. \quad (1.26)$$

Using the Bloch–Wigner identities [3, Eq. (3)],

$$D_2(z) = D_2\left(1 - \frac{1}{z}\right) = D_2\left(\frac{1}{1-z}\right) = -D_2\left(\frac{1}{z}\right) = -D_2(1-z) = -D_2\left(\frac{-z}{1-z}\right), \quad (1.27)$$

we note that

$$D_2\left(\frac{1-R}{1+R}\right) = -D_2\left(\frac{R+1}{1-R}\right), \quad D_2\left(\frac{R-1}{R+1}\right) = -D_2\left(\frac{R+1}{R-1}\right), \quad (1.28)$$

so that

$$\mathcal{P}(R) = \frac{2}{\pi} \left[D_2\left(\frac{R+1}{R-1}\right) - D_2\left(\frac{R+1}{1-R}\right) \right]. \quad (1.29)$$

We now use the five-term relation for the Bloch–Wigner,

$$D_2\left(\frac{1-x}{1-xy}\right) + D_2\left(\frac{1-y}{1-xy}\right) + D_2(1-xy) + D_2(x) + D_2(y) = 0, \quad (1.30)$$

with $x = R, y = -1$, to simplify the imaginary part of the period to the simple expression

$$\mathcal{P}(R) = \frac{2}{\pi} [D_2(-R) - D_2(R)]. \quad (1.31)$$

Since $D_2(\bar{z}) = -D_2(z)$, it follows that

$$\mathcal{P}(R) = 0 \quad \text{for real } R. \quad (1.32)$$

Repeating the analysis for U_2 yields identical exponents and roots, and thus the same result. For U_3 and U_4 , the change of variables is $t = \frac{-2iR(z+1)+z-1}{z-1}$, giving

$$\begin{aligned} (d_j) &= (1, 1, -1, -1), & (\alpha_j) &= \left(1, \frac{R-i}{R+i}, -1, \frac{i-R}{R+i}\right), \\ (e_j) &= (1, 1, -1, -1), & (\beta_j) &= \left(-1, \frac{R-i}{R+i}, 1, \frac{i-R}{R+i}\right). \end{aligned} \quad (1.33)$$

Hence,

$$\begin{aligned} \mathcal{P}(R) &= \frac{1}{\pi} \left[-D_2\left(\frac{R+i}{-R+i}\right) + D_2\left(\frac{R+i}{R-i}\right) + D_2\left(\frac{-R+i}{R+i}\right) - D_2\left(\frac{R-i}{R+i}\right) \right] \\ &= \frac{2}{\pi} \left[D_2\left(\frac{-R+i}{R+i}\right) - D_2\left(\frac{R-i}{R+i}\right) \right]. \end{aligned} \quad (1.34)$$

Applying the five term relation gives us

$$\tilde{\mathcal{P}}(R) = \frac{2}{\pi} [D_2(iR) - D_2(-iR)] \quad (1.35)$$

And this time we see that **But the formula should not apply**

$$\mathcal{P}(R) = 0 \quad \text{for imaginary } R. \quad (1.36)$$

1.3.3 Conifold period for general moduli

In fact, apart from the imaginary part of the period, [2, eq. (4.12)] also gives a formula for the full non-vanishing period at the conifold points, even when the curve is not real:

$$\begin{aligned} Q &= \frac{1}{2\pi} \sum_{j,k} d_j e_k \left[\text{Li}_2\left(\frac{\alpha_j}{\beta_k}\right) + (\log \alpha_j - \log \beta_k) \log\left(1 - \frac{\alpha_j}{\beta_k}\right) \right] \\ &\quad - \frac{1}{2\pi} \log y(0) (\log x(0) - \log x(\infty)) + \log y(0) \sum_j d_j \log \alpha_j \\ &\quad - \log x(\infty) \sum_k e_k \log \beta_k. \end{aligned} \quad (1.37)$$

where as before, $x(z)$ and $y(z)$ are rational parametrizations of the nodal curve in the original toric coordinates and the node corresponds to $z \rightarrow 0, \infty$. From now on, we work with the branches U_1 and U_2 but all the results can be generalized to U_3 and U_4 by taking $R \rightarrow iR$. Evaluating eq. (1.37) for the parametrization in eq. (1.22), we find the following after assuming $\text{Im } R > 0$ and simplifying logs

$$\begin{aligned} Q &= \frac{1}{\pi} \left(\text{Li}_2\left(\frac{-R-1}{R-1}\right) - \text{Li}_2\left(\frac{R+1}{R-1}\right) - \text{Li}_2\left(\frac{1-R}{R+1}\right) + \text{Li}_2\left(\frac{R-1}{R+1}\right) \right) \\ &\quad - \frac{2}{\pi} \log(R) \log(R-1) + i \log(R-1) + 2i \log(R) \\ &\quad + \frac{2}{\pi} \log(R) \log(R+1) - i \log(R+1). \end{aligned} \quad (1.38)$$

This expression can be simplified further thanks to various identities of the dilogarithm. We

use in particular the following identities [3]

$$\operatorname{Li}_2\left(\frac{1}{z}\right) = -\operatorname{Li}_2(z) - \frac{\pi^2}{6} - \frac{1}{2} \log(-z)^2, \quad (1.39)$$

$$\operatorname{Li}_2(1-z) = -\operatorname{Li}_2(z) + \frac{\pi^2}{6} - \log(z) \log(1-z). \quad (1.40)$$

We also have the so-called *five term relation*,

$$\begin{aligned} & \operatorname{Li}_2(x) + \operatorname{Li}_2(y) + \operatorname{Li}_2\left(\frac{x-1}{xy-1}\right) + \operatorname{Li}_2\left(\frac{y-1}{xy-1}\right) + \operatorname{Li}_2(1-xy) \\ &= \frac{\pi^2}{2} - \log(1-x) \log(x) - \log(1-y) \log(y) - \log\left(\frac{1-x}{1-xy}\right) \log\left(\frac{1-y}{1-xy}\right), \end{aligned} \quad (1.41)$$

which only holds near the vicinity of $x, y \in (0, 1)$ due to branch cuts. However, it is easy to find a generalized version of this identity for the domain we are interested in by starting with the identity in its twice differentiated form (where it reduces to a trivial algebraic identity) and integrating twice and fixing the integration constants.

Using the inversion identity eq. (1.39), we get

$$Q = -\frac{\pi}{2} - \frac{2}{\pi} \left[\operatorname{Li}_2\left(\frac{1-R}{R+1}\right) - \operatorname{Li}_2\left(\frac{R-1}{R+1}\right) - \log\left(\frac{1-R}{R+1}\right) \log(R) \right]. \quad (1.42)$$

We now invoke the five term relation with $(x, y) = (R, -1)$ and correct it with the procedure explained. After applying the reflection identity to some of the arguments eq. (1.40) and noting that $\operatorname{Li}_2(-1) = \frac{-\pi^2}{12}$, the identity reads simply

$$-\operatorname{Li}_2(-R) + \operatorname{Li}_2(R) + \operatorname{Li}_2\left(\frac{1-R}{R+1}\right) - \operatorname{Li}_2\left(\frac{R-1}{R+1}\right) = \frac{\pi^2}{4} - \log(R) \log\left(\frac{1-R}{R+1}\right). \quad (1.43)$$

Applying this to Q , we get our final expression

$$Q = -\pi - \frac{2}{\pi} [\operatorname{Li}_2(-R) - \operatorname{Li}_2(R)]. \quad (1.44)$$

2. Local $\mathbb{F}_1$

2.1 Nagell's algorithm for the mirror curve

We start from the equation of the local mirror of $\mathbb{F}_1$ defined as the elliptic curve,

$$x + y + \frac{1}{xy} + \frac{m}{x} = \frac{1}{u}, \quad (2.1)$$

where m and u are a priori free complex parameters. After multiplying by xy , we have

$$f(x, y) = x^2y + xy^2 - \frac{xy}{u} + my + 1 = 0. \quad (2.2)$$

We can eliminate the constant term by shifting $y \rightarrow y - \frac{1}{m}$,

$$f(x, y) = x^2y + xy^2 - \frac{x^2}{m} - \frac{(2u+m)xy}{um} + my + \frac{(m+u)x}{m^2u}. \quad (2.3)$$

We perform a series of affine transformations on x and y known as Nagell's algorithm as described in [4] to bring the curve into Weierstrass form. Note that in order for the curve to be elliptic, m and u cannot vanish simultaneously. Since we would be interested in the $m \rightarrow 0$ limit (Local $\mathbb{P}^2$), we switch $x \leftrightarrow y$ and assume generically that $m+u \neq 0$. We thus have

$$f(x, y) = x^2y + xy^2 - \frac{y^2}{m} - \frac{(2u+m)xy}{um} + mx + \frac{(m+u)y}{m^2u}. \quad (2.4)$$

Let us homogenize by writing $x = X/Z$, $y = Y/Z$ and clearing denominators. We get,

$$F(X, Y, Z) = F_3(X, Y) + F_2(X, Y)Z + F_1(X, Y)Z^2. \quad (2.5)$$

where

$$F_3 = X^2Y + XY^2, \quad (2.6)$$

$$F_2 = -\frac{(2u+m)}{um}XY - \frac{Y^2}{m}, \quad (2.7)$$

$$F_1 = mX + \frac{(m+u)}{m^2u}Y. \quad (2.8)$$

Note that the point $P = [0, 0, 1]$ is rational by construction (after eliminating the constant term). The tangent at P is given by $F_1 = mX + \frac{(m+u)}{m^2u}Y = 0$ and meets the curve at another point Q . We also define $e_i = F_i\left(\frac{(m+u)}{m^2u}, -m\right)$ ($e_1 = 0$ by definition) –

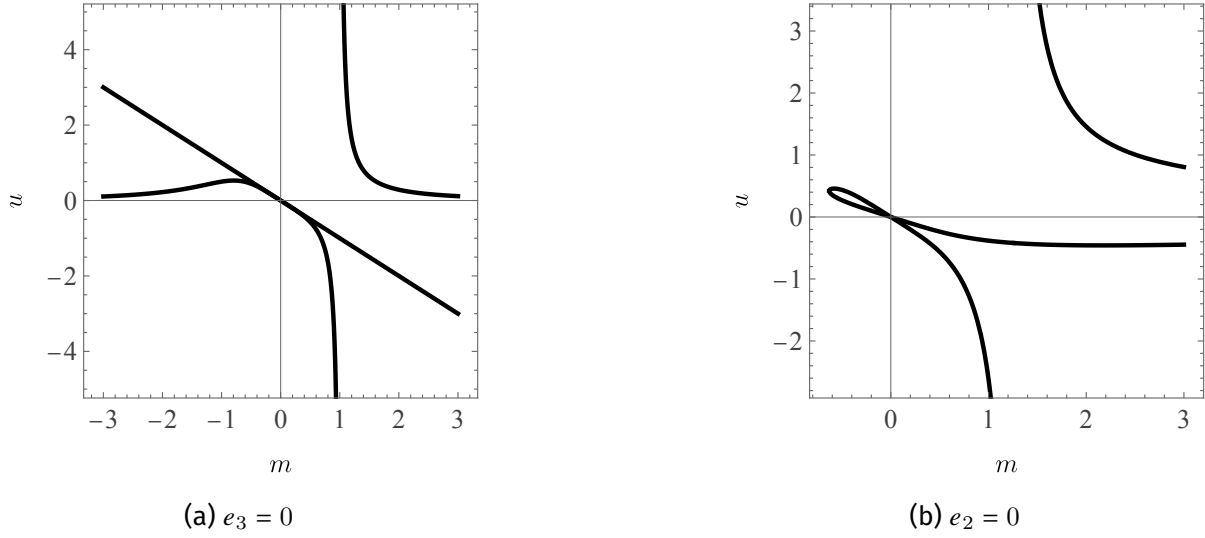
$$e_2 = \frac{2}{m^2} + \frac{3}{mu} - m + \frac{1}{u^2}, \quad (2.9)$$

$$e_3 = \frac{(m+u)\left(u - \frac{m+u}{m^3}\right)}{u^2}. \quad (2.10)$$

The coordinates of Q are given by

$$Q = \left[\frac{(m+u)(m^3u^2 - m^2 - 3mu - 2u^2)}{m^4u^3}, \right. \\ \left. -m^2 + \frac{m}{u^2} + \frac{2}{m} + \frac{3}{u}, \quad \frac{(m+u)\left(u - \frac{m+u}{m^3}\right)}{u^2} \right]. \quad (2.11)$$

For the curve to be elliptic, e_2 and e_3 cannot vanish simultaneously which is true provided m and u don't vanish simultaneously. On the real section of the $m-u$ plane, these conditions are plotted in fig. 2.

Figure 2: The loci where e_2, e_3 vanish.

Assuming generically that $e_3 \neq 0$, we make the coordinate change

$$X = X' + \frac{mu \left(\frac{2}{m^2} + \frac{3}{mu} - m + \frac{1}{u^2} \right)}{m + u - m^3u} Z' \quad (2.12)$$

$$Y = Y' - \frac{m^2 (m^3 (-u^2) + m^2 + 3mu + 2u^2)}{(m + u) (m + u - m^3u)} Z' \quad (2.13)$$

$$Z = Z' \quad (2.14)$$

which shifts Q to the origin, $[X', Y', Z'] = [0, 0, 1]$. We now shift back to affine coordinates $x' = X'/Z', y' = Y'/Z'$. The curve can now be written as

$$f'(x', y') = f'_3(x', y') + f'_2(x', y') + f'_1(x', y') \quad (2.15)$$

where

$$f'_3 = x'^2 y' + x' y'^2 \quad (2.16)$$

$$f'_2 = \frac{(m + u)^2}{mu (m + u - m^3u)} y'^2 + \left(\frac{2(m + u - m^3u)}{m(m + u)} + \frac{1}{u} \right) x' y' + \frac{m^2 (-(m^3 - 2)u^2 + m^2 + 3mu)}{(m + u) ((m^3 - 1)u - m)} x'^2 \quad (2.17)$$

$$f'_1 = \frac{(m + u) (2m^6 u^2 - m^5 - 5m^4 u - 4m^3 u^2 + m^2 + 2mu + u^2)}{m^2 u (m + u - m^3u)^2} y' + \frac{m}{u(m + u)^2 (m + u - m^3u)^2} (-m^5 - 6m^4 u + (m^3 - 15) m^3 u^2 - 3(m^6 - 3m^3 + 4) mu^4 + (6m^3 - 19) m^2 u^3 + (m^9 - 3m^6 + 4m^3 - 3) u^5) x' \quad (2.18)$$

We define $\phi_i = f'_i(1, t)$ and

$$\begin{aligned}
 \delta &= \phi_2^2 - 4\phi_1\phi_3 \\
 &= \frac{(m+u)^4}{m^2u^2(m+u-m^3u)^2}t^4 \\
 &\quad - \frac{2(m+u)(-(5m^3+1)u^2 - m(m^3+2)u - m^2 + 2m^2(m^3-2)u^3)}{mu^2(m+u-m^3u)^2}t^3 \\
 &\quad + \frac{1}{u^2(m+u-m^3u)^2}(2m(2m^3+1)u + m^2 + 2m(-2m^6+m^3+4)u^4 \\
 &\quad + (m^6+18m^3+1)u^2 + 2m^2(m^3+11)u^3)t^2 \\
 &\quad + \frac{2m}{u(m+u)(m+u-m^3u)^2}(m^4 + (m^3+4)m^3u + 2(-m^6+m^3+1)u^4 \\
 &\quad + (-3m^6+7m^3+6)mu^3 + (6m^3+7)m^2u^2)t \\
 &\quad + \frac{m^4(-(m^3-2)u^2 + m^2 + 3mu)^2}{(m+u)^2(m+u-m^3u)^2}
 \end{aligned} \tag{2.19}$$

Note that $\delta = 0$ whenever t is the slope of a tangent to the curve that passes through Q , in particular for $t = -\frac{m^3u}{m+u}$ (can be checked). We write $t = \frac{1}{\tau} - \frac{m^3u}{m+u}$ and define $\rho = \tau^4\delta$ which is a cubic in τ

$$\begin{aligned}
 \rho &= \frac{4m(m+u-m^3u)}{m+u}\tau^3 + \left(8m + \frac{1}{u^2}\right)\tau^2 + \frac{2(m+u)(2m^2u^2+m+u)}{mu^2(m+u-m^3u)}\tau \\
 &\quad + \frac{(m+u)^4}{m^2u^2(m+u-m^3u)^2}.
 \end{aligned} \tag{2.20}$$

Applying the final transformation,

$$\rho = \frac{(m+u)^2}{m^2(m+u-m^3u)^2}V^2, \tag{2.21}$$

$$\tau = \frac{(m+u)}{m(m+u-m^3u)}\left(U - \frac{1}{12}\left(8m + \frac{1}{u^2}\right)\right), \tag{2.22}$$

we obtain the final Weierstrass form,

$$V^2 = 4U^3 - g_2U + g_3, \tag{2.23}$$

with

$$g_2 = -\frac{4}{3}\left(-16m^2 + \frac{8m}{u^2} - \frac{1}{u^4} + \frac{24}{u}\right), \tag{2.24}$$

$$g_3 = -\frac{8}{27}\left(-64m^3 + \frac{48m^2}{u^2} - \frac{12m}{u^4} + \frac{144m}{u} + \frac{1}{u^6} - \frac{36}{u^3} + 216\right). \tag{2.25}$$

2.2 Periods as double Laurent series

From the GLSM charges defining the toric geometry,

$$\begin{array}{ccc|cc} 0 & 0 & 1 & -2 & -1 \\ 1 & 0 & 1 & 1 & 0 \\ -1 & -1 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 & 1 \end{array} \quad (2.26)$$

we see that the A-period of the mirror curve is given by the double sum

$$\begin{aligned} \omega_1 &= \sum_{k, \ell \geq 0} \frac{(k+2\ell)!}{(k!)^2(\ell!)^2} (-u)^{-k-2\ell} \lambda^{\ell-k} \\ &= \sum_{k, \ell \geq 0} \frac{(3k+2\ell)!}{(k!)^2(k+\ell!)^2} (-u)^{-3k-2\ell} \lambda^{\ell} \end{aligned} \quad (2.27)$$

where $z_1 = \frac{\lambda}{u^2}$, $z_2 = \frac{1}{u\lambda}$ are the standard complex structure parameters?
related to Marcos' parameters by $u_{\text{here}} = -\frac{1}{u_{\text{Marcos}}}$, $\lambda = m$?

where in the second line we shifted $\ell \mapsto \ell + k$, observing that terms with $k > \ell$ do not contribute in the sum in the first line. Indeed, we check that ω_1 is annihilated by the second order operator

$$\begin{aligned} \mathcal{D} &= (8\lambda u - 9) \left(16\lambda^3 + \lambda u^4 - u^3 - 8\lambda^2 u^2 + 36\lambda u - 27 \right) \theta_u^2 \\ &\quad + (-729 + 1512u\lambda - u^4\lambda - 936u^2\lambda^2 + 432\lambda^3 + 128u^3\lambda^3 - 512u\lambda^4) \theta_u \\ &\quad - 2(-243 + 486u\lambda - 282u^2\lambda^2 + 144\lambda^3 + 32u^3\lambda^3 - 192u\lambda^4) \end{aligned} \quad (2.28)$$

such that $\mathcal{D}_3 = u^{-3}\mathcal{D}\partial_u$. Using the Frobenius method, we find two logarithmic solutions

$$\begin{aligned} \omega_2 &= (-\log u - \log \lambda) \omega_1 + 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell)!}{(k!)^2(k!)^2(\ell-k)!} (H_{2\ell+k} + H_{\ell-k} - 2H_k) (-u)^{-k-2\ell} \lambda^{\ell-k} \\ &\quad + \sum_{0 \leq \ell < k} (-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell)!}{\ell! (k!)^2} u^{-k-2\ell} \lambda^\ell \\ \omega'_2 &= (-2\log u + \log \lambda) \omega_1 + 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell)!}{(k!)^2(k!)^2(\ell-k)!} (2H_{2\ell+k} - H_{\ell-k} - H_\ell) (-u)^{-k-2\ell} \lambda^{\ell-k} \\ &\quad - \sum_{0 \leq \ell < k} (-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell)!}{\ell! (k!)^2} u^{-k-2\ell} \lambda^\ell \end{aligned} \quad (2.29)$$

where $H_n = \sum_{k=1}^n (1/k)$ are the harmonic numbers. In fact, only the linear combination $2\omega_2 + 3\omega'_2$ is annihilated by $\mathcal{D}$, while ω_2 and ω'_2 satisfy it with a source term proportional to $9\lambda u^4 + 96\lambda^3 u^3$.

The CY periods satisfy $u\partial_u \varpi_2 = \omega_1$, $u\partial_u \varpi_3 = \frac{1}{4}(2\omega_2 + 3\omega'_2)$. Hence they are obtained by

integrating once with respect to u ,

$$\begin{aligned}
 \varpi_2 &= -\log u + \sum_{3k+2\ell>0} \frac{(3k+2\ell-1)!}{(k!)^2(\ell!)(k+\ell)!} (-u)^{-3k-2\ell} \lambda^\ell \\
 \varpi_3 &= \log^2 u + (2\log u - \frac{1}{4}\log \lambda) \varpi_2 + \frac{1}{8}\log^2 \lambda \\
 &\quad - 2 \sum_{-2\ell < k \leq \ell} \frac{(k+2\ell-1)!}{(k!)^2(k)!(\ell-k)!} (2H_{2\ell+k} - H_{\ell-k} - H_\ell) (-u)^{-k-2\ell} \lambda^{\ell-k} \\
 &\quad + \sum_{0 \leq \ell < k} (-1)^\ell \frac{\Gamma(k-\ell)(k+2\ell-1)!}{\ell! (k!)^2} u^{-k-2\ell} \lambda^\ell - 2 \sum_{k+2\ell>0} \frac{(k+2\ell-1)!}{(k+2\ell)(k!)^2(k)!(\ell-k)!} u^{-k-2\ell} \lambda^\ell
 \end{aligned} \tag{2.30}$$

To be checked numerically

2.3 Convergence domain of Frobenius solution as a function of m

Consider the series

$$f(u, m) = \log u + \sum_{n \geq 2} \sum_{3k+2\ell=n} \frac{(n-1)!}{(k+\ell)! (k!)^2 \ell!} m^\ell u^n. \tag{2.31}$$

For fixed complex parameter m , this is a series in the complex variable u . We are interested in the radius of convergence $R(m)$ of the power-series part around $u = 0$.

The logarithmic term $\log u$ contributes a singularity at $u = 0$, but it does not affect the radius of convergence of the power-series part. Thus we focus on

$$g(u, m) := \sum_{n \geq 2} a_n(m) u^n, \quad a_n(m) := \sum_{3k+2\ell=n} \frac{(n-1)!}{(k+\ell)! (k!)^2 \ell!} m^\ell. \tag{2.32}$$

The radius of convergence for fixed m is

$$\frac{1}{R(m)} = \limsup_{n \rightarrow \infty} |a_n(m)|^{1/n}. \tag{2.33}$$

2.3.1 Rewriting as a double series

It is convenient to write the power series directly as a sum over (k, ℓ) :

$$g(u, m) = \sum_{k, \ell \geq 0} \frac{(3k+2\ell-1)!}{(k+\ell)! (k!)^2 \ell!} m^\ell u^{3k+2\ell}. \tag{2.34}$$

For fixed m, u , the general term in absolute value is

$$T_{k, \ell}(R, M) := \left| \frac{(3k+2\ell-1)!}{(k+\ell)! (k!)^2 \ell!} \right| R^{3k+2\ell} M^\ell, \quad R = |u|, \quad M = |m|. \tag{2.35}$$

The radius $R(m)$ is determined by the exponential growth of the coefficients in $n := 3k + 2\ell$, i.e. by $\limsup |a_n(m)|^{1/n}$.

2.3.2 Saddle-point analysis via Stirling

For large k, ℓ with fixed ratio, we approximate factorials using Stirling's formula:

$$n! \sim n^n e^{-n} \sqrt{2\pi n}. \quad (2.36)$$

Introduce the scaling

$$k = \alpha n, \quad \ell = \beta n, \quad n := 3k + 2\ell, \quad (2.37)$$

so that

$$3\alpha + 2\beta = 1, \quad \alpha, \beta \geq 0. \quad (2.38)$$

A single term in the inner sum defining $a_n(m)$ then behaves like

$$\log \left| \frac{(n-1)!}{(k+\ell)! (k!)^2 \ell!} m^\ell \right| \sim n \phi(\alpha, \beta; M), \quad (2.39)$$

for some function ϕ that can be extracted from the Stirling approximation. Using the constraint $3\alpha + 2\beta = 1$, we can solve

$$\alpha = \frac{1-2\beta}{3}, \quad 0 \leq \beta \leq \frac{1}{2}. \quad (2.40)$$

In terms of β alone, one finds

$$\phi(\beta; M) = \log \left(\frac{M^\beta}{\alpha^{2\alpha} \beta^\beta (\alpha + \beta)^{\alpha+\beta}} \right), \quad \alpha = \frac{1-2\beta}{3}, \quad (2.41)$$

up to subleading (subexponential in n) terms.

For large n , the sum over ℓ (equivalently over β) is dominated by the value of β that maximizes $\phi(\beta; M)$ for fixed $M = |m|$. Hence

$$\limsup_{n \rightarrow \infty} |a_n(m)|^{1/n} = \exp \left(\sup_{0 \leq \beta \leq 1/2} \phi(\beta; M) \right). \quad (2.42)$$

Denote

$$\log \Lambda(M) := \sup_{0 \leq \beta \leq 1/2} \phi(\beta; M). \quad (2.43)$$

Then the radius of convergence is

$$R(m) = \frac{1}{\Lambda(|m|)}. \quad (2.44)$$

2.3.3 Stationary point and parametric description

The dominant contribution comes from a saddle point $\beta_*(M)$ where

$$\frac{d\phi}{d\beta}(\beta_*(M); M) = 0. \quad (2.45)$$

A straightforward differentiation and simplification give

$$\frac{d\phi}{d\beta} = \log\left(\frac{M}{3\beta(\beta+1)^{1/3}}\right) + \frac{4}{3}\log(1-2\beta). \quad (2.46)$$

The saddle condition thus reads

$$\log\left(\frac{M}{3\beta(\beta+1)^{1/3}}\right) = -\frac{4}{3}\log(1-2\beta), \quad (2.47)$$

or equivalently

$$M = 3\beta(\beta+1)^{1/3}(1-2\beta)^{-4/3}. \quad (2.48)$$

For each $M = |m| > 0$, there is (under suitable regularity assumptions) a unique $\beta_*(M) \in (0, 1/2)$ solving this equation.

Evaluating the growth factor $\Lambda(M)$ at the saddle, one finds that the corresponding exponential growth rate can be written as

$$\Lambda(M) = E(\beta_*(M)), \quad (2.49)$$

where

$$E(\beta) = 3 \frac{(1-2\beta)^{-2/3}}{(1+\beta)^{1/3}}. \quad (2.50)$$

Therefore, the radius of convergence is

$$R(m) = \frac{1}{\Lambda(|m|)} = \frac{1}{3} (1-2\beta_*)^{2/3} (1+\beta_*)^{1/3}, \quad (2.51)$$

with $\beta_* = \beta_*(|m|)$ determined implicitly by (2.48).

In other words, we can give a parametric description of the boundary of the convergence domain in the $(|u|, |m|)$ -plane. Introducing a real parameter $\beta \in (0, 1/2)$, we set

$$\boxed{\begin{aligned} M(\beta) &= 3\beta(1+\beta)^{1/3}(1-2\beta)^{-4/3}, \\ R(\beta) &= \frac{1}{3}(1-2\beta)^{2/3}(1+\beta)^{1/3}, \end{aligned}} \quad (2.52)$$

so that for fixed m with $M = |m|$, one determines the corresponding β via $M(\beta) = |m|$, and then the radius of convergence is $R(m) = R(\beta)$.

2.3.4 Special limits

The case $m = 0$. Taking the limit $M \rightarrow 0$ corresponds to $\beta \rightarrow 0$. From $M(\beta) \sim 3\beta$ as $\beta \rightarrow 0$, we have $\beta \sim M/3$, while

$$R(\beta) \xrightarrow{\beta \rightarrow 0} \frac{1}{3}. \quad (2.53)$$

Thus

$$R(0) = \frac{1}{3}. \quad (2.54)$$

Large $|m|$. For $|m| \rightarrow \infty$, the parameter β approaches $1/2$ from below. From the parametric form one can extract the asymptotic behaviour

$$R(m) \sim C |m|^{-1/2}, \quad |m| \rightarrow \infty, \quad (2.55)$$

for some explicit constant $C > 0$ (which can be worked out from the leading behaviour of $M(\beta)$ and $R(\beta)$ near $\beta = 1/2$).

2.3.5 Summary

For each fixed value of the parameter $m \in \mathbb{C}$, the power-series part of $f(u, m)$ in u has a finite radius of convergence $R(m)$ given implicitly by a saddle-point analysis:

- Introduce $M = |m|$ and a parameter $\beta \in (0, 1/2)$.
- The pair (M, R) is parametrized by

$$M(\beta) = 3\beta(1+\beta)^{1/3}(1-2\beta)^{-4/3}, \quad R(\beta) = \frac{1}{3}(1-2\beta)^{2/3}(1+\beta)^{1/3}. \quad (2.56)$$

- Given m , solve $M(\beta) = |m|$ for β and then $R(m) = R(\beta)$.

The logarithmic term $\log u$ does not affect the radius of convergence of the power-series part, though it introduces a singularity at $u = 0$ itself.

2.4 Rational change of coordinates and the Kerr-Doran prescription

Nagell's algorithm provides a systematic procedure to convert the mirror curve of local $\mathbb{F}_1$ into Weierstrass form, but it does so at the price of introducing a transcendental change of variables. A significantly simpler rational transformation, originally presented in [5], is given by

$$x = \frac{36u^2(m+3u) + 6X + \sqrt{2}Y - 9}{6u(12mu^2 + 2X - 3)}, \quad y = \frac{36u^2}{-12mu^2 - 2X + 3}. \quad (2.57)$$

In these coordinates the mirror curve eq. (2.1) assumes Weierstrass form,

$$Y^2 = 4X^3 - g_2X - g_3, \quad (2.58)$$

with

$$\begin{aligned} g_2 &= 27 \left(16m^2u^4 - 8mu^2 - 24u^3 + 1 \right), \\ g_3 &= 27 \left(64m^3u^6 - 48m^2u^4 - 144mu^5 + 12mu^2 - 216u^6 + 36u^3 - 1 \right). \end{aligned} \quad (2.59)$$

These expressions agree with eq. (2.24) up to an overall normalization, which can be absorbed by rescaling X and Y . The discriminant takes the form

$$\Delta \propto u^8 \delta(u, m), \quad \delta(u, m) = 16m^3u^4 - 8m^2u^2 - 36mu^3 + m - 27u^4 + u, \quad (2.60)$$

showing that for fixed m the curve generically develops four nodal degenerations corresponding to the roots of $\delta(u, m) = 0$. Among these solutions we single out the branch $u_*(m)$ of smallest absolute value; this root is the one governing, for example, the convergence of the large-radius expansion of the periods, and will therefore play a distinguished role in the analysis.

To treat all nodal branches uniformly we follow the formalism of [5]. At the nodal locus the curve can be written as

$$Y^2 = 4(X - X_0)^2(X - X_1). \quad (2.61)$$

Eliminating the X^2 term enforces $X_1 = -2X_0$. Matching coefficients with the Weierstrass form then yields

$$X_0 = -\frac{3g_3}{2g_2}, \quad (2.62)$$

evaluated on the discriminant locus $\delta(u, m) = 0$. A convenient rational parametrization of eq. (2.61) is

$$X = \frac{1}{2} \left(t^2 - 4X_0 \right), \quad Y = \frac{t(t^2 - 6X_0)}{\sqrt{2}}. \quad (2.63)$$

Expressed back in the original toric coordinates, this leads to

$$\begin{aligned} x &= \frac{36u^2(m + 3u) + 3(t^2 - 4X_0) + t(t^2 - 6X_0) - 9}{6u(12mu^2 + t^2 - 4X_0 - 3)}, \\ y &= \frac{36u^2}{-12mu^2 - t^2 + 4X_0 + 3}. \end{aligned} \quad (2.64)$$

Introducing the notation

$$t_1^2 = -12mu^2 + 4X_0 + 3, \quad (2.65)$$

the denominators become $t^2 - t_1^2 = (t - t_1)(t + t_1)$. On the discriminant locus one may verify that the numerator of x factorizes as

$$36u^2(m + 3u) + 3(t^2 - 4X_0) + t(t^2 - 6X_0) - 9 = (t - t_1)^2(t + 2t_1 + 3), \quad (2.66)$$

which leads to the one-parameter family

$$x = \frac{(t - t_1)(t + 2t_1 + 3)}{6u(t + t_1)}, \quad y = -\frac{36u^2}{t^2 - t_1^2}. \quad (2.67)$$

To apply the Kerr–Doran prescription we shift the nodes to 0 and ∞ . From eq. (2.63) the node occurs at $t = \pm\sqrt{6X_0}$, leading to the coordinate

$$z = \frac{t - \sqrt{6X_0}}{t + \sqrt{6X_0}}. \quad (2.68)$$

In terms of z , the parametrization eq. (2.67) takes the form

$$x = \alpha \frac{(z-a)(z-b)}{(z-c)(z-1)}, \quad y = \beta \frac{(z-1)^2}{(z-b)(z-c)}, \quad (2.69)$$

where

$$\begin{aligned} b &= \frac{t_1 - \sqrt{6X_0}}{t_1 + \sqrt{6X_0}}, & a &= \frac{3 + 2t_1 + \sqrt{6X_0}}{3 + 2t_1 - \sqrt{6X_0}} = \frac{1}{b^2}, & c &= \frac{t_1 + \sqrt{6X_0}}{t_1 - \sqrt{6X_0}} = \frac{1}{b}, \\ \alpha &= \frac{t_1 + 3}{6u} = \left(\sqrt{b} + \frac{1}{\sqrt{b}} \right)^{-\frac{4}{3}}, & \beta &= -\frac{18u^2}{t_1^2 + 3t_1} = \left(\sqrt{b} + \frac{1}{\sqrt{b}} \right)^{\frac{2}{3}}. \end{aligned} \quad (2.70)$$

Using the general expression for the imaginary part of the period,

$$\mathcal{P} = \frac{1}{2\pi} \sum_{j,k} d_j e_k D_2 \left(\frac{\alpha_j}{\beta_k} \right), \quad (2.71)$$

one finds

$$\mathcal{P}(b) = \frac{1}{2\pi} \left[D_2(b^3) - 4D_2(b^2) + 5D_2(b) \right]. \quad (2.72)$$

The full non-vanishing period follows from

$$\begin{aligned} \mathcal{Q} &= \frac{1}{2\pi} \sum_{j,k} d_j e_k \left[\text{Li}_2 \left(\frac{\alpha_j}{\beta_k} \right) + (\log \alpha_j - \log \beta_k) \log \left(1 - \frac{\alpha_j}{\beta_k} \right) \right] \\ &\quad - \frac{1}{2\pi} \log y(0) (\log x(0) - \log x(\infty)) + \log y(0) \sum_j d_j \log \alpha_j - \log x(\infty) \sum_k e_k \log \beta_k, \end{aligned} \quad (2.73)$$

giving the explicit expression

$$\begin{aligned} \mathcal{Q}(b) &= \frac{1}{2\pi} \left[\text{Li}_2(b^3) - 4 \text{Li}_2(b^2) + 5 \text{Li}_2(b) \right] \\ &\quad + \frac{1}{4\pi} \log b \left(6 \log(b^2 + b + 1) + \log b - 16 \log(b + 1) - 4i\pi \right). \end{aligned} \quad (2.74)$$

1. A minor mismatch remains in the real part of the period when evaluated at $(m, u) = (1, u_*(1))$.
2. In contrast, the discrepancy in the imaginary part noted in [5] for $(m, u) = (2, u_*(2))$ is fully resolved by the above formula.
3. The expressions for α and β in terms of b given in eq. (2.70) apply only to the branch corresponding to $m = 1$ with $u = u_*(1) \approx 0.26318$.

2.5 Large Volume Central Charge

We take $\mathbb{F}_1$ to be the blow-up of $\mathbb{P}^2$. We define a basis of compact curve classes on local $\mathbb{F}_1$. Let C be the exceptional class, $C^2 = -1$ in the cup product. Let H be the hyperplane class, $H^2 = 1$ and $H.C = 0$. We consider compactly supported coherent sheaves E on local $\mathbb{F}_1$ with Chern vector $\gamma(E) = [r(E), c_1(E), \text{ch}_2(E)]$, where r is the rank of the sheaf on $\mathbb{F}_1$, $c_1(E)$ is the first Chern class

2.6 To organize

Sending $\log(\tilde{u}) \rightarrow \log(-\tilde{u})$ in the solutions ϖ_2 and ϖ_3 of the PF system transforms (T, T_D) as follows

$$T \rightarrow T - \frac{1}{2}, \quad T_D \rightarrow T_D - 4T - \frac{m}{2} + 1. \quad (2.75)$$

which at the level of charges corresponds to

$$(r, d_F, d_B, \text{ch}_2) \rightarrow (r, d_F + \frac{3}{2}r, d_B + r, \text{ch}_2 + d_F + \frac{1}{2}d_B + r) \quad (2.76)$$

Whereas, under the large volume monodromies sending $\log(\tilde{u}) \rightarrow \log(\tilde{u}) + 2\pi i k$, we get

$$T \rightarrow T - k, \quad T_D \rightarrow T_D - 8kT - km + 4k^2. \quad (2.77)$$

2.7 Large Volume Scattering

We take for the large volume central charge of an object E of chern Character $\gamma = (r, d_F, d_B, \text{ch}_2)$, the function

$$Z_{\text{LV}}(\gamma) = -r \left(4T^2 + mT - \frac{m^2}{2} \right) + d_F(2T + m) + d_B(T - m) - \text{ch}_2. \quad (2.78)$$

2.7.1 Walls of Marginal Stability

Walls of marginal stability between two objects of charge γ and γ' occur when their central charges are aligned, $\arg Z_{\text{LV}}(\gamma) = \arg Z_{\text{LV}}(\gamma') \pmod{2\pi}$. It is possible to show that for fixed real m , walls of marginal stability are circles in the T -plane. The center is at

$$T_0 = \frac{\text{ch}_2' r - \text{ch}_2 r' - d_B m r' + d_B' m r - d_F' m r + d_F m r'}{\langle \gamma, \gamma' \rangle} \in \mathbb{R} \quad (2.79)$$

on the real axis, while the radius R can be written as

$$R^2 = T_0^2 + \frac{T_0(-d_B' - 2d_F' + m r')}{4r'} + \frac{2\text{ch}_2' - m(-2d_B' + 2d_F' + m r')}{8r'} \quad (2.80)$$

2.7.2 Rays

Given a phase $\psi \in [0, \pi)$, the ray corresponding to an object E of charge γ is given by

$$\text{Ray equation: } \mathcal{F}(\gamma, T, m, \psi) = \operatorname{Re}(e^{-i\psi} Z_{LV}(\gamma)) = 0. \quad (2.81)$$

and $\operatorname{Im}(e^{-i\psi} Z_{LV}(\gamma)) > 0$. Of course, this ray may or may not be populated over the entire T plane but the equation above is the support of the ray whenever it exists (also for the object of charge $-\gamma$). A direct evaluation of $\mathcal{F}$ for the central charge in eq. (2.78) shows that the rays are rectangular hyperbolas in the plane of $T = s + it$. The center of the hyperbola is at

$$T_0 = \frac{1}{8} \left[\frac{(d_B + 2d_F)}{r} - m \right], \quad (2.82)$$

After performing a rotation by $\frac{\psi}{2}$ and shifting the center, defining $T' = e^{-i\frac{\psi}{2}}(T - T_0) = s' + it'$ and $m = m_1 + im_2$, the ray equation looks like

$$\begin{aligned} s'^2 - t'^2 = & \frac{-2r(8\text{ch}_2 + (9d_B - 6d_F)m_1) + (d_B + 2d_F)^2 + 9r^2(m_1^2 - m_2^2)}{64r^2} \cos \psi \\ & + \frac{3m_2(-3d_B + 2d_F + 3m_1r)}{32r} \sin \psi. \end{aligned} \quad (2.83)$$

2.7.3 Rays in x-y coordinates

We define coordinates

$$x = \frac{\operatorname{Re}(e^{-i\psi} T)}{\cos \psi}, \quad y = -\frac{\operatorname{Re}(e^{-i\psi} T_D)}{\cos \psi}, \quad \mu = \frac{\operatorname{Re}(e^{-i\psi} m)}{\cos \psi}, \quad (2.84)$$

which makes the rays into straight lines

$$x(d_B + 2d_F) + ry - \text{ch}_2 + \mu(d_F - d_B) = 0 \quad (2.85)$$

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